SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1714**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Priyanka.G /192524124 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

Resolution Algorithm Implementation

Student 31: Priyanka G | Register Number: 192524124

Question 1 - Sports Achievement Award

Knowledge Base

1. If a player scores above 90 percent, then the player wins a gold medal.
2. Divya scored above 90 percent.

Goal

Prove that **Divya wins a gold medal.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Job Interview Outcome

Knowledge Base

1. If an applicant passes the interview, then the applicant is hired.
2. Farhan passed the interview.

Goal

Prove that **Farhan is hired.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Attendance Credit

Knowledge Base

1. If a student attends all classes, then the student gets attendance credit.
2. Meena attended all classes.

Goal

Prove that **Meena gets attendance credit.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Car Rental Eligibility

Knowledge Base

1. If a driver has a valid license, then the driver can rent a car.
2. Suresh has a valid license.

Goal

Prove that **Suresh can rent a car.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access.** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
5. Explain each resolution step and write the final conclusion.

DATE: 13/08/26 | CSA1714 ARTIFICIAL INTELLIGENCE | NAME: PRIYANKA G

CO 3 ASSESSMENT TOOL 3 | REG.NO: 192524124.

RESOLUTION ALGORITHM IMPLEMENTATION

1) SPORTS ACHIEVEMENT AWARD

(i) Propositional logic Representation.

Let: P = Divya scores above 90 percent.

G = Divya wins a gold medal.

Knowledge Base:

- If player scores above 90 percent, then the player wins a gold medal. $P \rightarrow G$.

Divya scored above 90 percent. P.

Goal: G

(ii) Convert to CNF and Negate the Goal.

Implication: $P \rightarrow G$

using: $A \rightarrow B \equiv \neg A \vee B$

$\therefore \quad \neg P \vee G$

Fact: P

Negation of the goal: $\neg G$

(iii) Resolution Algorithm

Resolve C1 and C2 on P: $(\neg P \vee G)$, P

Resolving P and $\neg P$: G

New Clause: C4: G

Now resolve C4 with C3: G, $\neg G$

$\therefore \boxed{\square}$

(iv) Resolution Proof.

step	Clauses	Resolvent
{1}	$\neg P \vee G$, P	G

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Step	Clauses	Resolvent
{2}	$C_1, \neg G$	$\boxed{\square}$

Final conclusion: Since the resolution process derives the empty clause $\square$, the negation of the goal is contradictory with knowledge Base.

$\boxed{\therefore \text{ Divya wins a gold medal.}}$

2) JOB INTERVIEW OUTCOME :

(i) Propositional Logic Representation

Let:

- P = Farhan passes the interview.
- H = Farhan is hired.

Knowledge Base :

If an applicant passes the interview, then the applicant is hired. $P \to H$.

Farhan passed the interview. P

Goal: H

(ii) CNF conversion and Negated Goal :

Convert : $P \to H$

to CNF : $\neg P \vee H$

Fact : P

Negate the goal : $\neg H$

Clauses : $C_1 = \neg P \vee H$

$C_2 = P$

$C_3 = \neg H$

(iii) Resolution Steps :

Resolve C_1 and C_2 : $\neg P \vee H$, P

$\therefore$ H

New clause : C_4 : H

Resolve C_4 and C_3 : $H, \neg H$

$\therefore \boxed{\square}$

(iv) Resolution Table

Step	Clauses	Resolvent
{1}	$\neg P \vee H, P$	H
{2}	$H, \neg H$	$\boxed{\square}$

Final Conclusion:

The empty clause is obtained. Hence, the negated goal is false and the original goal is logically proved.

$\boxed{\therefore \text{ Farhan is hired.}}$

3) ATTENDANCE CREDIT

(i) Propositional Logic Representation:

Let:

- A = Meena attends all classes.
- C = Meena gets attendance credit.

Knowledge Base:

- If a student attends all classes, then the student gets attendance credit. $A \to C$.
- Meena attended all classes. A.

Goal: C

(ii) CNF Conversion and Negated Goal:

Convert the Implication: $A \to C$

$\therefore \neg A \vee C$

Fact: A

Negated Goal: $\neg C$

Clauses:

- C_1 : $\neg A \vee C$
- C_2 : A
- C_3 : $\neg C$

(iii) Resolution Steps

Resolve C1 and C2: $\neg A \vee C$, A

$\therefore$ C

New Clause : C_4 : C

New Resolve C_4 with C_3 : C, $\neg C$

$\therefore$ $\boxed{\square}$

(iv) Resolution Table

Step	Clauses	Resolvent
{1}	$\neg A \vee C$, A	C
{2}	C, $\neg C$	$\boxed{\square}$

Final Conclusion: The empty clause is derived, proving that the negation of the goal is inconsistent with the knowledge base.

$\therefore$ Meena gets attendance credit.

4) CAR RENTAL ELIGIBILITY

(i) Propositional Logic Representation

Let :

- L = Suresh has a valid license.
- R = Suresh can rent a car.

Knowledge Base :

- If a driver has a valid license, then the driver can rent a car. $L \rightarrow R$.
- Suresh has a valid license. L

Goal : R

(ii) CNF Conversion and Negated Goal

Convert : $L \rightarrow R$

Using : $A \rightarrow B \equiv \neg A \vee B$

We obtain : $\neg L \vee R$

Fact : L

Negated Goal : $\neg R$

Clauses :

C1 : $\neg L \vee R$

C2 : L

C3 : $\neg R$

(iii) Resolution Steps:

Resolve C1 and C2 : $\neg L \vee R$, L

The complementary literals L and $\neg L$ cancel.

$\therefore R$

New clause : $C4 = R$

Now resolve C4 and C3 : $R, \neg R$

$\therefore \boxed{\square}$

(iv) Resolution Table :

Step	Clauses	Resolvent
{1}	$\neg L \vee R, L$	R
{2}	$R, \neg R$	$\boxed{\square}$

Final conclusion: Since the empty clause is obtained, the goal is logically proved.

$\therefore$ Suresh can rent a car.

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5) ACCESS CONTROL SYSTEM:

(i) Propositional Logic Representation:

Let :

- P = The user enters the correct password.
- A = The ~~user~~ is authenticated.
- G = The user is grated access

Knowledge Base :

Statement 1 : If the user enters the correct password, then the user is authenticated, $P \rightarrow A$.

Statement 2 : If the user authenticated, then the user is grated access. $A \rightarrow G$.

Statement 3: The user entered the correct password. P

Goal: G

(ii) Convert all implications into CNF

First implication: $P \to A$

Using: $P \to A \equiv \neg P \vee A$

$\therefore \quad \neg P \vee A$

Second Implication: $A \to G$

$\therefore \neg A \vee G$

Fact: P

Thus, the Knowledge-base clauses are:

- $C_1: \neg P \vee A$
- $C_2: \neg A \vee G$
- $C_3: P$

(iii) Negate the Goal

Goal: G

For resolution by contradiction, negate the goal: $\neg G$.

$\therefore$, The complete resolution set is:

$C_1 = \neg P \vee A, \quad C_2 = \neg A \vee G, \quad C_3 = P, \quad C_4 = \neg G$

(iv) Resolution Table

Step	Parent Clauses	Resolution	New Clause
{1}	$C_1: \neg P \vee A$, $C_3: P$	Resolve $P, \neg P$	$C_5: A$
{2}	$C_2: \neg A \vee G$, $C_5: A$	Resolve $A, \neg A$	$C_6: G$
{3}	$C_6: G$, $C_4: \neg G$	Resolve $G, \neg G$	$\boxed{\square}$

Final Conclusion

Since empty clause $\square$ is derived, the negation of the goal leads to a contraction. Hence, the original goal is locally entailed by the Knowledge Base.

$\boxed{\therefore \text{ The user is granted access}}$